

# C5 - MULTIMODAL RECOGNITION

## Task 2: Object Segmentation

### Team 1:

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MASTER IN  
COMPUTER VISION  
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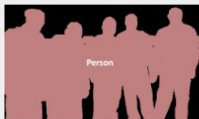
# Setup & Introduction

## Object Segmentation

**Instance Segmentation:** Segments each object separately, assigning a **unique mask to every individual instance of a class** (each car or pedestrian).



**Semantic Segmentation:** Segments the image by class, **assigning the same label to all pixels belonging to the same category** without distinguishing individual objects.



**Note:** In all experiments we perform **instance segmentation**, except for the optional task.

## Goals

- **Primary:** The goal is to learn how to do segmentation with different prompting strategies. We test **pretrained SAM** on KITTI-MOTS, use **Grounded SAM** with text prompts for text-guided segmentation, and finally fine-tune SAM.
- **Domain Shift:** Evaluate pretrained and finetuned SAM on the *DeART* dataset
- **Optional:** Semantic segmentation

## Frameworks & Models

- **Hugging Face** ([Models Doc](#)):
  - SAM  Meta
  - Grounding DINO (**IDEA Research**)



**Part of the  
Hugging Face course!**



- **We first outline our experimental framework before detailing task results.**
- **Github Repository:** <https://github.com/valy3124/C5-Team1>

# Model: Segment Anything Model (SAM)

**Note:** For all the experiments, we use the "facebook/sam-vit-base" model.

**Def:** SAM is a **foundation model** designed to **segment any object in an image without task-specific training**, using prompts such as points, bounding boxes, or masks, making it useful for interactive segmentation and annotation

## Image Encoder

1

- Vision Transformer (ViT) backbone.
- Extracts deep **image embeddings** from the input image.
- Runs **once per image** and serves as the foundational feature extractor.

## Prompt Encoder

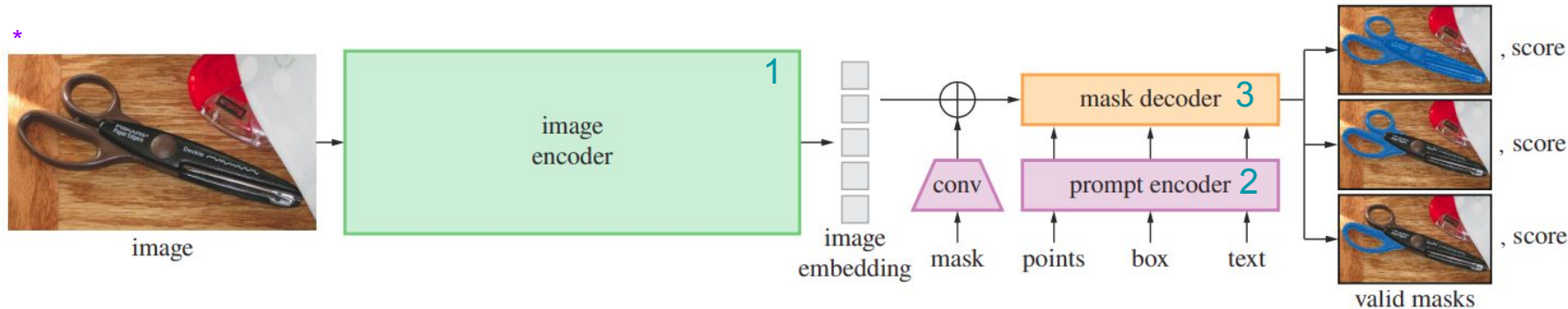
2

- Encoder that translates **user instructions into embeddings**.
- Allows SAM to be a **promptable model** by choosing what to segment based on specific information.
- Handles **Sparse** prompts (Points, Bounding Boxes, Text) and **Dense** prompts (Masks).

## Mask Decoder

3

- Final prediction head.
- Combines the image embeddings and the prompt embeddings to predict the **final valid segmentation masks**.
- Generates the masks alongside **confidence scores** for each prediction.



\* Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollar, Ross Girshick; Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023, pp. 4015-4026 [https://openaccess.thecvf.com/content/ICCV2023/html/Kirillov\\_Segment\\_Anything\\_ICCV\\_2023\\_paper.html](https://openaccess.thecvf.com/content/ICCV2023/html/Kirillov_Segment_Anything_ICCV_2023_paper.html)

# Model: Grounding DINO

**Note:** All experiments use  
"IDEA-Research/grounding-dino-tiny"

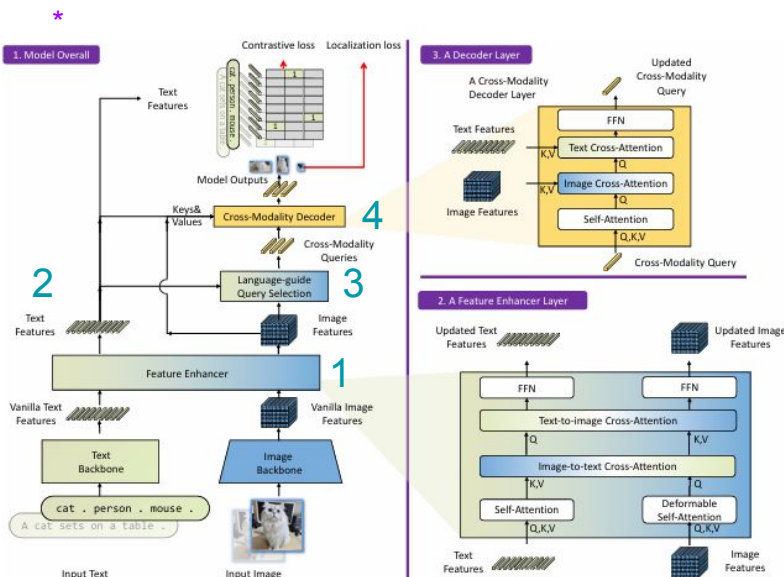
**Def:** Grounding DINO is an **open-vocabulary object detection model** that detects objects in an image based on **text prompts**.

## Feature Extraction and Enhancer

- Extract **image features** with an image backbone.
- Extract **text features** with a text backbone.
- Use **multi-scale image features** from different backbone layers.
- Apply **self-attention** to enhance image and text features separately.
- Use **cross-attention (image $\leftrightarrow$ text)** to fuse and align both modalities.

## Sub-Sentence Level Text Feature

- **Sentence-level** encodes the whole sentence into one feature but **loses fine-grained details**.
- **Word-level** encodes multiple category names but **creates dependencies between categories**.
- The proposed approach extracts **sub-sentence features** to keep **more detailed semantic information**.



## Language-Guided Query Selection

- The model receives **image features** and **text features**.
- Computes **similarity between image tokens and text tokens**.
- Selects the **top 900** image features most relevant to the text (**decoder queries**)

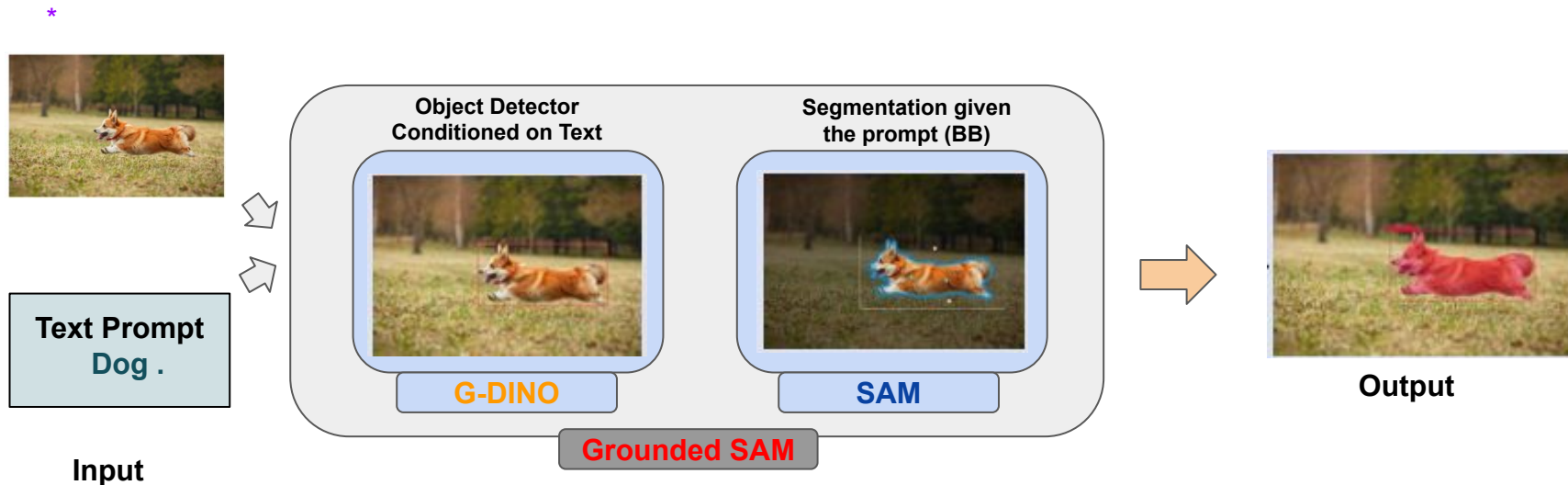
## Cross-Modality Decoder

- Each query goes through **self-attention** to refine query interactions (remove duplicates).
- Uses **image cross-attention** to integrate visual features.
- Uses **text cross-attention** to inject language information.
- Improves **alignment between text prompts and detected objects**.



# Model: Grounded SAM

**Def:** Grounded SAM is a pipeline that combines **Grounding DINO** and **SAM** to perform **text-guided object segmentation**.



\* T. Ren *et al.*, "Grounded SAM: Assembling Open-World Models for Diverse Visual Tasks," *arXiv preprint arXiv:2401.14159*, 2024.

# Dataset: KITTI-MOTS

**2 classes:** Cars and Pedestrians (**Huge class imbalance !!!**)

**Number of sequences:** 21 sequences: 12 for training and 9 for validation (**split based on the paper \***)

We think this imbalance is natural due to the urban driving setting, where vehicles appear more frequently and persist longer in the scene than pedestrians.



## Model Development Phase

### Train

Used for the initial training phase (**80% of the full training data: 9 seqs**)

### Dev

Used for hyperparameter tuning and model selection without touching the final validation set (**20% of the full training data: 3 seqs**)

## Final Training and Evaluation Phase

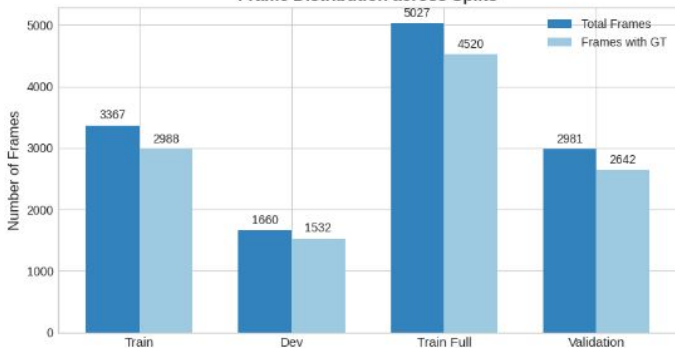
### Full Train

Used for the **final training stage**, combining Train and Train-Dev once the best hyperparameters are selected, to maximize available training data.

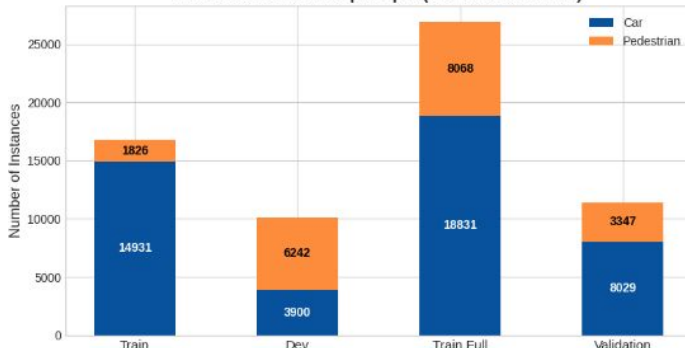
### Validation

**Reserved strictly for final evaluation.** It provides an unbiased estimate of the model's generalization performance.

Frame Distribution across Splits



Instance Distribution per Split (Car vs. Pedestrian)



## Key Observation

- **Most frames** contain ground truth annotations.
- **Strong class imbalance is observed:** *Car* instances significantly outnumber *Pedestrians* in every split.
- The dataset is overall **car-dominant**, which may bias the model toward better car detection

\* Voigtlaender, P., Krähenbühl, P., Luiten, J., Leal-Taixé, L., & Schiele, B. (2019). MOTs: Multi-Object Tracking and Segmentation. In "Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)" (pp. 7942–7951).

# Task a) Point Prompting Strategies

Our **goal is to segment cars and pedestrians** using the **pretrained SAM model** on the **KITTI MOTS dataset** (model size mentioned in [slide 3](#)). Since SAM requires prompts to generate segmentation masks, the **choice of prompting strategy can strongly influence the results**. Therefore, we **qualitatively** evaluate **three different strategies**: **uniform grid points**, **SIFT keypoints**, and the **center of the ground-truth bounding boxes** to understand how prompt location affects segmentation quality.



## Grid

Prompt points are **uniformly distributed** across the image following a regular grid pattern (**10 points per side**). This approach does not rely on prior information about object locations and instead **explores the entire image to segment possible objects**.



## SIFT

We use the SIFT (Scale-Invariant Feature Transform) algorithm to detect **distinctive keypoints** in the image based. These keypoints are then used as prompts, focusing on **the most informative regions** of the image \*



## Center GT Bounding Box

We use the **geometric center of the ground-truth bounding boxes (cars and pedestrians)** as the prompt point. This simulates an ideal scenario where **the model receives a perfect indication** of where the interest object is located.

**\*Note:** Default max\_points is set to 60 to counteract SIFT's tendency to cluster on highly textured regions, ensuring broader spatial coverage of the scene

# Task a) Qualitative Examples (Cars)

Ground Truth (GT)



Ground Truth (GT)



Prompt

Prediction

Grid



Prompt

Prediction



SIFT



Center  
GT BB



**Note:** After model prediction, **Non-Maximum Suppression (NMS)** with an IoU threshold of 0.8 is applied. This removes overlapping masks and results in cleaner final predictions. Only for Grid and SIFT strategies

**Note:** Since SAM outputs **three candidate masks per prompt**, only the mask with the **highest confidence** score is selected.

**Note:** The discussion about this **will be presented in the next slide**, where we analyze the different qualitative examples. The discussion about these results will be presented in the next slide, where we compare qualitative examples in car scenes and pedestrian scenes.



# Task a) Qualitative Examples (Pedestrians)

Ground Truth (GT)



Prompt

Prediction

Grid



SIFT



Center  
GT BB



Ground Truth (GT)



Prompt

Prediction



## Point Prompting Observations

- **Unlocalized point prompts (Grid, SIFT)** often produce good masks for **large regions** such as *trees, road, or buildings*. However, when a point falls on a **local part of a larger object with a different texture**, SAM may segment **that specific region instead of the whole object**. For example, a point on a **window** may produce a mask for the window rather than the entire building.
- **Localized prompts (Center BB)** provide **more reliable results** because the point lies **inside the interest object**. However, **object type affects performance** since segmentation tends to work **better for cars than for pedestrians**. Cars have **more compact shapes and homogeneous appearance**, while pedestrians have **thin structures and pose variability**, making them harder to segment. This can be seen in the qualitative examples, where SAM sometimes segments only specific parts of the person, such as the pants or the shirt, instead of the full pedestrian.

# Task a) Metric for Quantifying the Qualitative Examples

**Motivation:** After presenting qualitative examples, we now **quantitatively evaluate** the behavior of each prompting strategy using segmentation metrics. This allows us to objectively compare the different strategies and **identify their strengths, weaknesses, and overall reliability**.

## Mask Area

**Def:** Measures the number of pixels predicted as part of the object.

**Why:** Important to detect over-segmentation and relation with SAM confidence.

$$A = \sum_{x=1}^W \sum_{y=1}^H M(x, y)$$

**where**

**$M(x, y)$  is the mask**

## Fragmentation

**Def:** Number of connected components in a mask.

**Why:** Indicates whether the segmentation is compact or broken into many pieces.

$$F(m) = C(m)$$

**where**

$$C(m) = |\{c_1, c_2, \dots, c_k\}|$$

## SAM Confidence

**Def:** SAM's predicted mask quality score for each candidate mask (SAM outputs three masks per prompt).

**Why:** Used to analyze how reliable this confidence score is depending on which mask is selected (top-1, top-2, top-3) and the prompting strategy used.

## Fragmentation Rate

**Def:** Fraction of masks with more than one connected component.

$$FR = \frac{\sum_{i=1}^N \mathbf{1}(F(m_i) > 1)}{N}$$

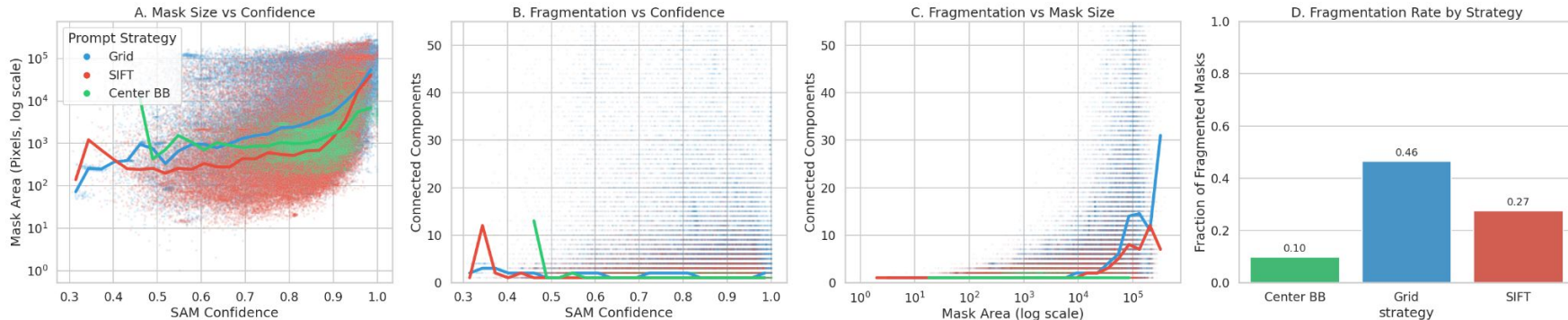
**Why:** Helps compare which prompting strategies produce more stable masks.

**where**

**$\mathbf{1}(F(m_i) > 1)$  means if there is more than 1 connected component**

# Task a) Quantifying Qualitative Examples

## Segmentation Behavior



## Key Observations

- **Mask size vs Confidence:** Higher SAM confidence scores **are generally associated with larger segmented areas**. Very small masks often correspond to ambiguous sub-parts or background noise, which naturally **lowers the model's certainty**.
- **Fragmentation vs Confidence:** Fragmentation **does not lower confidence**. SAM's internal confidence score predicts the overall IoU, **not topological perfection**. Therefore, the model **can remain highly confident** even when a mask is shattered into dozens of pieces.
- **Fragmentation vs Mask Size:** Larger masks tend to produce **exponentially more connected components**. As a mask expands, it inevitably **encounters occlusions or complex boundaries**.
- **Fragmentation by strategy:** Semantic prompting (**Center BB**) ensures solid masks by targeting the object's core. Blind point generation (**Grid**) frequently hits edges or shadows, **creating a huge volume of noise**.
- **Action:** **Center BB** is the most stable prompting strategy and should be preferred, as it produces the **lowest fragmentation rate and more compact masks by focusing prompts on the object's core**. It will be used strategy in the finetuning experiments.



# Task a) Mask Analysis

**Motivation:** In the previous slides, we always selected the mask with the highest confidence. However, SAM actually returns **three candidate masks** for each prompt. In this analysis, we examine **how these masks differ (\*)**

Ground Truth (GT)



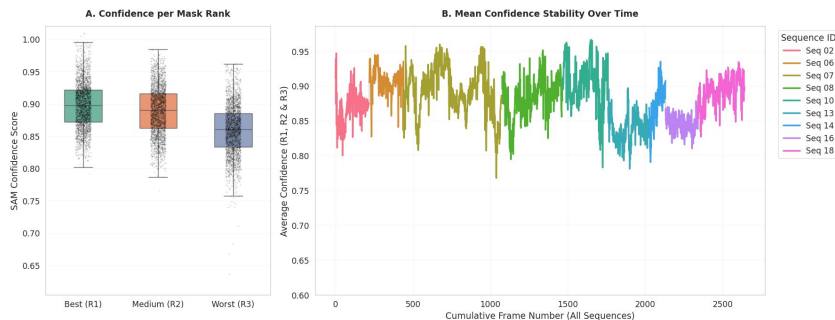
Top 1 Mask



Top 2 Mask



Top 3 Mask



➔ Sequence 16

## Key Observations

- **Small differences between the top masks:** From the qualitative examples, **Mask 1 and Mask 2 are usually very similar**, while **Mask 3 tends to differ more noticeably**. This behavior is also reflected in the **confidence distribution plot**.
- **SAM produces consistently high confidence masks:** The average confidence across the three masks remains high across all sequences. However, a small decrease in confidence is observed in **Sequence 16**, which corresponds to scenes with **many pedestrians, more objects to segment, and frequent occlusions**, increasing the overall segmentation difficulty.
- **Action:** Selecting the highest-confidence mask (Top 1) is a **reliable strategy**

## Task b) Inference using different text prompts

**Motivation:** Our goal is to segment objects using **text prompts instead of predefined dataset labels**. This is possible thanks to **Grounded SAM** (explained in [slide 5](#)). We use different **text prompts** to evaluate the model's ability to understand not only standard object categories (**prompt 1**) but also **object parts** (**prompt 2**) and **scene-related concepts** (**prompt 3**).

### Prompt 1:

Pedestrians . Cars



### Predictions



### Prompt 2:

wheels . car window . license plate



### Prompt 3:

road . traffic light . building . sky



### Key Observations

**Grounded SAM** struggles to segment pedestrians (**case 1**) with standard prompts and shows **inconsistent results for object parts** such as **wheels, license plates, or car windows** (**case 2**). In contrast, **stuff categories** like **road** or **sky** are segmented more reliably. **These observations motivate the following experiments, where we further investigate the causes behind these behaviors.**

# Task b) Analyzing Failure Cases in Grounded SAM

**Case 1:** Why does the model fail when using *pedestrian* as a prompt?

**\* Note:** Remember that the original class on KITTI MOTTS is pedestrian not person.

**Example 1**

**Prompt:** Pedestrian



**Prompt:** Person



**Example 2**

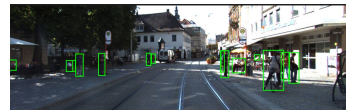
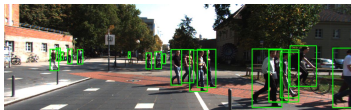
**Prompt:** Pedestrian



**Prompt:** Person



**Experiment:** Let's analyze the prompt that Grounding DINO provides to SAM



## Key Observations

**Grounding DINO** recognizes **person** better than **pedestrian**, producing more accurate bounding boxes. Since SAM relies on these boxes as prompts, segmentation quality depends on this detection step. **But why does this happen?** We explore this question further through additional cases in the next slide.

## Task b) Analyzing Failure Cases in Grounded SAM

**Case 2:** Why does the model fail with prompts such as wheels, license plate, and car window?

**Prompt:** wheels . car window . license plate ➡ **Bad Results**



**Prompt:** traffic light . stop sign . helmet . backpack . umbrella . shoe ➡ **Good Results**

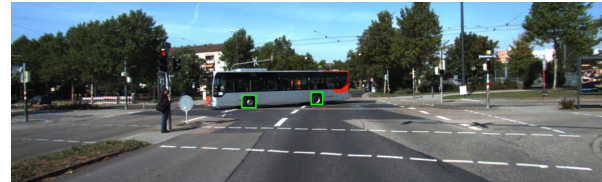


**Experiment:** Same experiment as in the previous slide. Let's analyze the prompt that Grounding DINO provides to SAM (**next slide**)

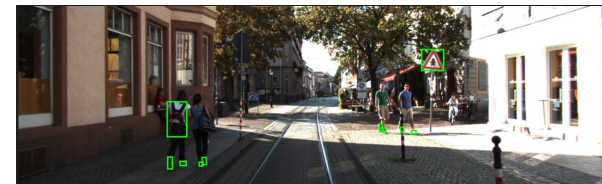


# Task b) Analyzing Failure Cases in Grounded SAM

**Prompt:** wheels . car window . license plate → **Bad Detections**



**Prompt:** traffic light . stop sign . helmet . backpack . umbrella . shoe → **Good Detections**



## Key Observations

According to the Grounding DINO paper, the **Grounding DINO Tiny** model was mainly pretrained on **Objects365** and **GoldG**. In Objects365, classes such as **person** are explicitly present, while some more specific concepts relevant to our prompts (e.g., *pedestrian*, *wheel*, *car window*, *license plate*) are not defined as object categories. GoldG, on the other hand, is a **visual grounding dataset**, where objects are associated with text descriptions rather than fixed object classes. As a result, prompts corresponding to well-represented concepts (e.g., **person**) tend to produce stronger detections, while more specific or uncommon prompts may receive lower confidence scores.

**Reasoning:** Some valid detections may receive **low confidence scores** due to the prompt not matching well with the concepts seen during pretraining. Since Grounding DINO uses a **box threshold of 0.35**, these detections may be filtered out, reducing the number of predicted objects.

**Experiment:** In the next slide, we **vary the box threshold** and analyze how it affects the number of detections and the final segmentation results. This experiment allows us to verify whether the threshold is filtering valid detections due to low confidence scores.

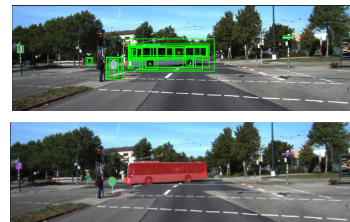
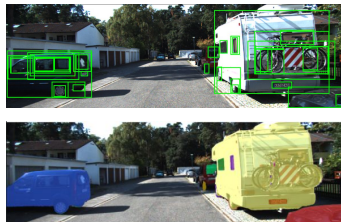
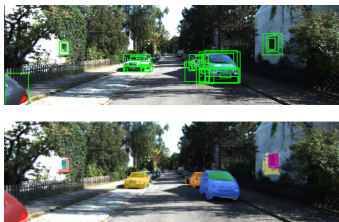


## Task b) Analyzing Failure Cases in Grounded SAM

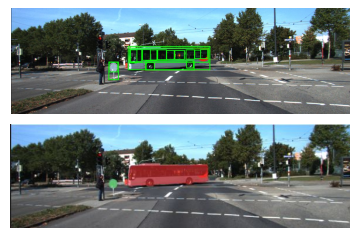
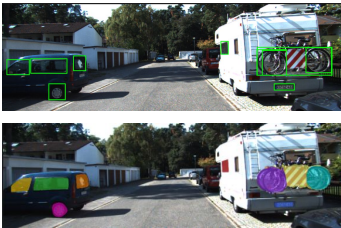
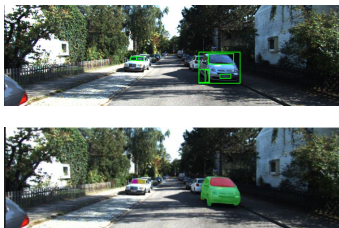
**Experiment:** Let's vary the **Grounding DINO box threshold** to analyze whether the default value we were using (**0.35**) was too restrictive and caused valid detections to be filtered out. We evaluate several threshold values and observe how the quality of bounding boxes change.

**Prompt:** wheels . car window . license plate

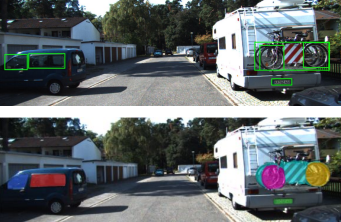
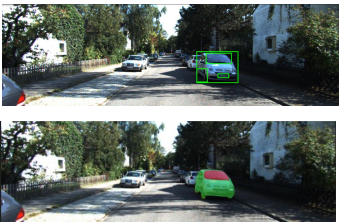
BB Threshold: 0.10



BB Threshold: 0.20



BB Threshold: 0.25



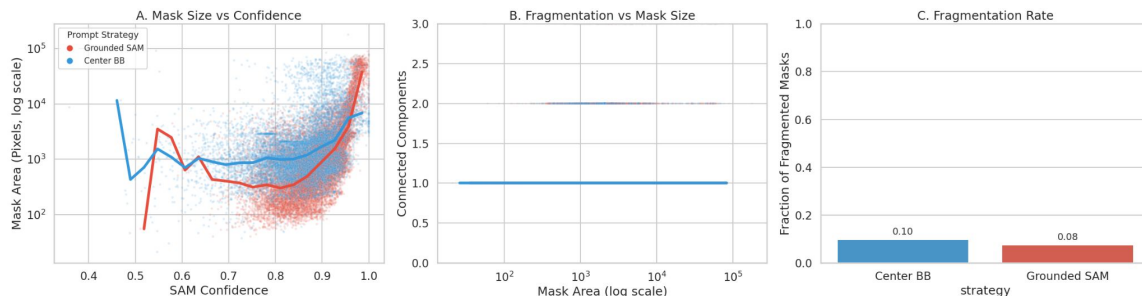
# Task b) Conclusion of the experiments + Quantifying Qualitative examples

## Conclusions

- As shown in the previous slide, when we significantly reduce the **Grounding DINO box threshold**, some of the desired objects are detected, but not all of them. As the threshold increases, the detections become more reliable and precision improves, although the number of detected objects decreases considerably. This indicates that **the threshold itself is not the main issue**, since even with very low values the model still fails to detect many relevant objects. Therefore, **our initial hypothesis about the threshold is not supported**. A more plausible explanation is that the pretrained model has seen **very few examples of these specific concepts**, which limits its detection capability even when the confidence threshold is relaxed.
- On the other hand, we observed that the prompts that work best for segmenting the **KITTI-MOTS classes** (*pedestrian* and *car*) are “**person**” and “**car**”, as these correspond to classes present during Grounding DINO’s pretraining. These prompts will be used in the next analysis, where we will perform a **quantitative evaluation**, followed by **qualitative examples** in the subsequent slide. Moreover, **these prompts will be used in the remaining experiments of the project, since they have shown the best performance on KITTI-MOTS**.

## Quantifying Qualitative Examples: Grounded SAM vs Center BB

Segmentation Behavior: Grounded SAM vs Center BB



## Key Observations

- Prompt information**: Center BB uses only the box center as a prompt, while Grounded SAM uses the full bounding box. **The richer spatial guidance explains the slightly lower or similar fragmentation rate.**
- Mask area vs confidence**: The same behavior observed before appears here as well, confirming that the confidence–area relationship is **consistent**.

## Task b) Examples using the correct KITTI-MOTS

**Motivation:** So far, we have not shown examples comparing the segmentation results obtained using the two prompts that produce good segmentation for KITTI-MOTS with the ground truth. For this reason, in this slide we present some examples using the final prompt.

**Prompt:** Person . Cars

**Ground Truth (GT)**



**Prediction**



### Key Observations

- Using prompts that provide better results ("Person", "Car"), which correspond to classes present during DINO's pretraining, the model produces generally accurate segmentations that align well with the visible objects. However, these prompts **do not perfectly match the KITTI-MOTS taxonomy**. In particular, the prompt "person" **includes all human instances, while the dataset only annotates pedestrians**, excluding cases such as cyclists. As a result, some detections that are correct for the prompt **are not present in the ground truth** and are therefore counted as **false positives during evaluation**. Additionally, in scenes with **occlusions**, the model may struggle to fully segment the objects, as illustrated in the second example.
- This highlights a limitation of using open-vocabulary detectors with datasets that follow a different class definition. A possible solution would be to adapt the prompts to better match the dataset taxonomy or fine-tune the detector on the target classes.

# Task c) Metric Description

## Instance Segmentation Metrics

### Mean Average Precision (mAP) (Main Metric)

- Average of the **Average Precision (AP)** computed over multiple **IoU thresholds (0.50–0.95)**, summarizing the overall detection or segmentation performance.

$$mAP = \frac{1}{10} \sum_{t=0.50}^{0.95} AP_{IoU=t}$$

### AP Small

- **Small Objects:**  $area < 32^2$

### AP Medium

- **Medium Objects:**  $32^2 < area < 96^2$

### AP Large

- **Small Objects:**  $area > 96^2$

### AP Car

- Average Precision computed considering **only the car class**, measuring how accurately the model detects and segments cars.

### AP Person

- Average Precision computed considering **only the person class**, measuring how accurately the model detects and segments persons.

### F1 Score

- In task g) we also included this metric to **evaluate the balance between precision and recall**, providing a single measure of detection/segmentation quality that considers both false positives and false negatives.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

## Semantic Segmentation Metrics

### Mean Intersection Over Union (mIoU)

- Measures the average overlap between the predicted segmentation masks and the ground truth masks across all classes.
- Penalize FP predictions
- Favors region smoothness and does not evaluate boundary accuracy
- **Formula:**

$$IoU = \frac{|Prediction \cap GroundTruth|}{|Prediction \cup GroundTruth|}$$

$$mIoU = \frac{1}{C} \sum_{i=1}^C IoU_i$$

where C is the number of classes and IoU<sub>i</sub> the metric computes for class i.

# Task c) Quantitative Comparison Between Our Best Detection Model and a Pretrained Detector

**Motivation:** We use **pretrained SAM** with bounding boxes as prompts to **segment cars and pedestrians in the KITTI-MOTS dataset**. The bounding boxes are generated by **two models**: a **YOLOv10-B pretrained on COCO** and a **YOLOv10-B fine-tuned on KITTI-MOTS**. SAM then produces the segmentation masks **in a zero-shot manner** using these boxes as prompts. This allows us to perform both **qualitative and quantitative comparisons** to analyze how the detection quality affects the final segmentation.

## Setup

- **Pretrained Detector Model:**
  - YOLO V10-B (pretrained on COCO)
  - mAP: **61.5** (in detection)
- **Our Detector Model:**
  - YOLO V10-B fine tuned on KITTI MOTTS.
  - mAP: **65.5** (in detection)

## Results

- **Pretrained Detector Model + SAM:**
  - mAP (validation): **52.6**
- **Finetuned Detector Model + SAM:**
  - mAP (validation): **54.2**
- **Improvement: + 1.6**

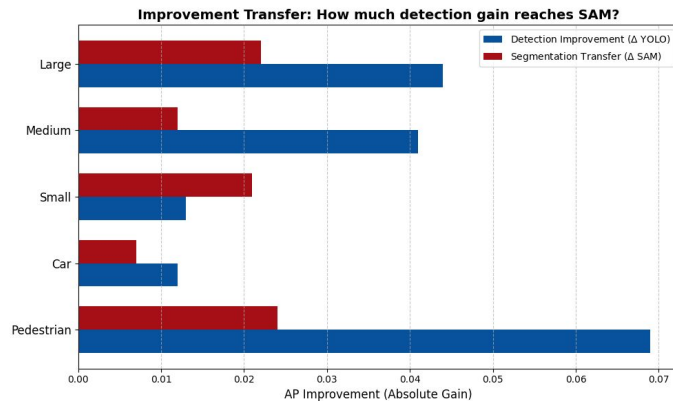
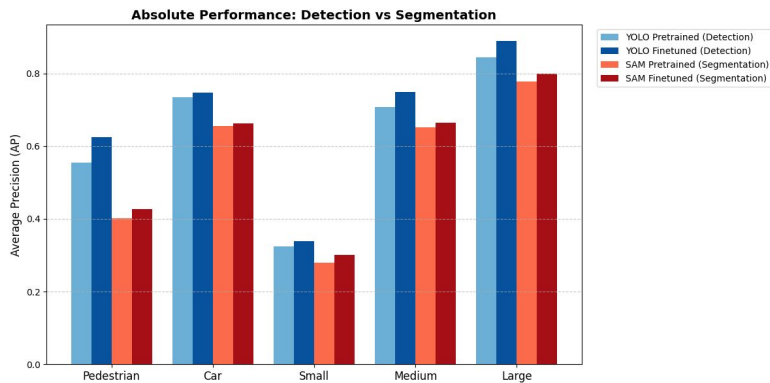
## Reasoning

- The +1.6 mAP segmentation improvement is driven by the fine-tuned detector providing **more reliable prompts and fewer false positives**. However, this gain is limited: **once SAM receives a good enough box, giving it slightly tighter coordinates doesn't noticeably alter its final mask**

**Experiment:** While **fine-tuning notably improves detection accuracy**, the data shows that these spatial gains **do not transfer proportionally to the final SAM segmentation masks**. In the **following slide**, we break down this discrepancy **by analyzing the performance transfer across specific object classes and size categories**.



# Task c) Quantitative Comparison Between Our Best Detection Model and a Pretrained Detector



## Conclusions

- **Difficulties with pedestrians:** Human bodies are tricky to outline. Even when our fine-tuned model gives SAM a perfect box, SAM still struggles to trace detailed edges like arms and legs without specific training.
- **Already good boxes:** For medium and large objects, the original boxes were already good enough for SAM. Making a box slightly tighter improves our detection scores a lot, but it doesn't teach SAM how to draw a better mask.
- **Action:** Using the **fine-tuned detector is still the best choice**. Even though the mask improvement isn't 1-to-1, giving SAM more accurate boxes provides a better starting point, resulting in cleaner masks overall (+1.6 mAP).

# Task c) Qualitative Comparison Between Our Best Detection Model and a Pretrained Detector

Ground Truth (GT)



Ground Truth (GT)



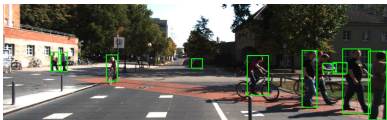
Input Prompts

Predictions

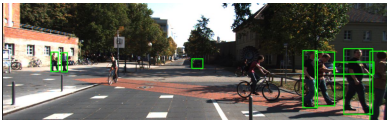
Input Prompts

Predictions

Pretrained  
YOLO V10-B +  
SAM



Finetuned  
YOLO V10-B +  
SAM



## Key Observations

- **Domain Adaptation & Box Limits: In the Left Example** The fine-tuned detector successfully ignores cyclists (reducing false positives) and produces slightly tighter bounding boxes. Although the difference is subtle, it can be observed in the **rightmost mask**. However, SAM generates almost identical segmentation masks in both cases, indicating that the original boxes were already sufficiently accurate to guide the segmentation. In the **Right Example** both detectors provide solid boxes, yet SAM still includes the backpack in the mask. Better bounding boxes cannot fix SAM's internal struggle with complex object boundaries. It also adapts to the new dataset, reducing incorrect masks.

# Task d) Quantitative Comparison Between Grounding DINO Prompts and Our Best Detector Prompts

**Motivation:** We compare two ways of generating prompts for SAM to segment cars and pedestrians in the KITTI-MOTS dataset. In the **first approach**, **bounding boxes are produced by our best detector (YOLOv10-B fine-tuned on KITTI-MOTS)**. In the **second approach**, bounding boxes are generated using **Grounding DINO with text prompts** within the Grounded SAM pipeline. SAM then produces the segmentation masks in a **zero-shot manner** using these boxes as prompts. This allows us to quantitatively and qualitatively analyze how different prompt sources (dataset-specific detection vs text-guided detection) affect the final segmentation performance.

## Setup

- **Segmentation with our detector::**
  - YOLO V10-B fine tuned on KITTI MOTTS + SAM
- **Grounded SAM:**
  - Grounding DINO + SAM.
  - **Prompt used:** Person . Car

## Results

- **Finetuned Detector Model + SAM:**
  - mAP (validation): **54.2**
- **Grounded SAM:**
  - mAP (validation): **55.3**
- **Improvement: + 1.1**

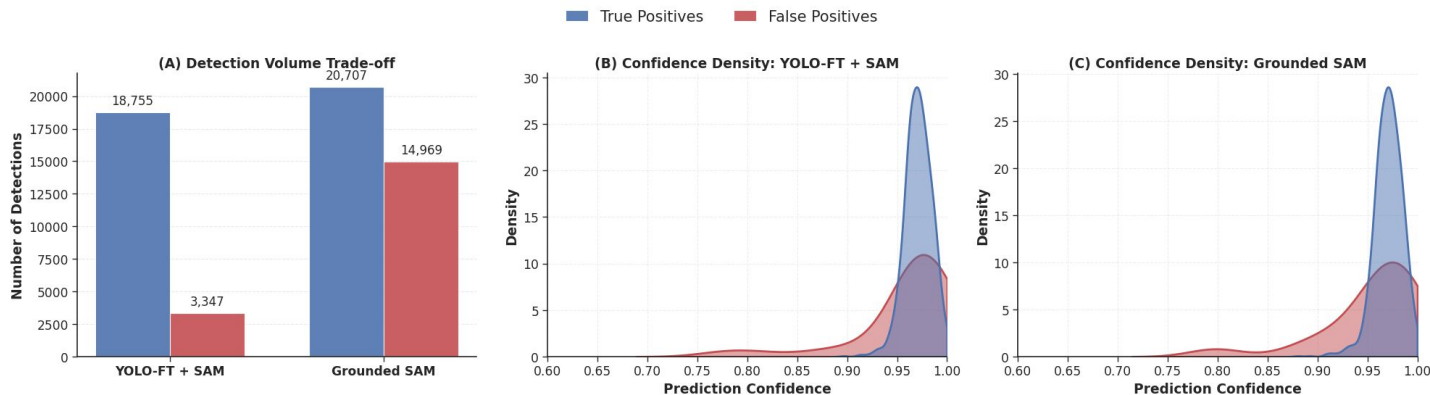
## Reasoning

- Grounding DINO's large-scale pre-training **may provide higher recall**, which can slightly improve segmentation performance. However, this +1.1 mAP advantage **may come at the cost of a higher false positive rate**. Many of these detections likely have **low confidence and therefore have limited impact on the mAP metric.** (\*) **Experiment to proof this**

**Experiment:** To validate this hypothesis, we will conduct **in the next slide** a detailed error analysis comparing both pipelines. We will evaluate the overall **Recall** to confirm if Grounding DINO finds more hidden objects, count the total **False Positives (FPs)** to measure the noise, and analyze the **confidence distribution of FPs** to verify whether most of them occur at low confidence levels, which would explain their limited impact on the final mAP.

# Task d) Quantitative Comparison Between Grounding DINO Prompts and Our Best Detector Prompts

Performance and Confidence Distributions: Finetuned YOLO vs Grounded SAM



## Conclusions

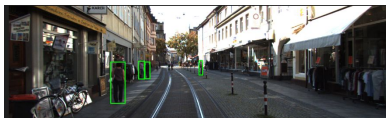
- **Grounded SAM increases recall but also introduces more false positives.** However, most false positive appear at lower confidence scores, so they are ranked later in the precision-recall curve and penalize the AP only marginally.
- **Action: Grounded SAM achieves higher segmentation performance.** It obtains **55.3 mAP**, outperforming the pipeline using YOLOv10-B fine-tuned on KITTI-MOTS + SAM (**54.2 mAP**).

# Task d) Qualitative Comparison Between Grounding DINO Prompts and Our Best Detector Prompts

Ground Truth (GT)



Input Prompts



Predictions



Ground Truth (GT)



Input Prompts



Predictions



Finetuned  
YOLO V10-B +  
SAM

Grounded  
SAM



## Key Observations

In both examples, **Grounded SAM detects more objects** for both classes (*person* and *car*), increasing the number of true positives compared to the pipeline using the fine-tuned YOLO detector. However, this higher recall also introduces **more false positives**, mainly because some distant or partially visible objects are not annotated in the KITTI-MOTS ground truth.



# Task e) Approach: Fine-Tuning & Prompt Strategy

## The Rationale & Constants:

- **Paradigm Shift:** Prioritizing prompt exploration (Points, Text, Mixed) over exhaustive hyperparameter tuning due to compute limits and prior C3 experience.
- **Fixed Parameters:** Shared across all fine-tuning runs to isolate the impact of the prompt type.

LR: 5e-5 | Epochs: 15 | Grad Clip: 1.0

Optimizer: AdamW | Batch size: 8

## Engineering for Batch Training

To enable efficient batched fine-tuning for SAM, we implemented custom PyTorch workarounds to handle variable-length ground truth data:

- **Dynamic Batching:** Padded prompt sequences with placeholders to create uniform tensor shapes.
- **Native Batched Upscaling:** Replicated SAM's output post-processing natively to preserve gradient flow across batches.
- **Gradient-Aware Loss:** Sliced prediction tensors to compute DiceBCE loss strictly on valid objects, ignoring the padding.

## Augmentation Search (Baseline)\*

- **Input:** Bounding Boxes (Bbox)
- **Objective:** Grid search to define the most robust augmentations pipeline, **optimized for peak mAP** (detailed in [Slide 20](#)).
- **Data Split:** Custom **train/dev split** (detailed in [Slide 6](#)) used to tune hyperparameters and compare models without touching validation data
- **Output:** Locks in the single best augmentation strategy for the next phases



## Prompt-Decoder Fine-Tuning Exploration

- **Approach:** Retrain the Prompt-Decoder using the locked augmentation strategy across different prompt modalities.
- **Data Split:** Trained on **full train split**.



Bbox\*

Point\*

Text\*

Mixed\*



## Final Evaluation

- **Data Split:** Evaluated strictly on the **Official Validation set**.
- **Analysis:** Assessed both quantitative (COCO metrics) and qualitative results.
- **Comparison:** Directly compared fine-tuned models against pre-trained baselines across all prompt modalities.

## Task e) Fine-Tuning: Prompt Modality Strategies

### Bbox Prompts

- **Input:** **Ground Truth (GT)** bounding boxes.
- **Rationale:** Used to evaluate segmentation performance given **perfect localization**. *(Due to time constraints, we skipped using Week 1's predicted boxes to focus strictly on the prompt-decoder's **upper-bound capabilities**).*
- **Implementation:** Direct **1:1 mapping** between GT boxes and GT masks. Serves as our primary baseline.

### Point Prompts

- **Input:** A **single positive spatial point** per object.
- **Implementation:** Extracted automatically by calculating the geometric center of each GT bounding box, a strategy validated earlier in **Task a)** ([Slide 11](#))

### Mixed Prompting (*Dynamic Training*)

- **Training Strategy:** Prompt modality **changes randomly per batch**. The model must learn to decode boxes, points, and text outputs **simultaneously**.
- **Validation Strategy:** Evaluated the single final model exhaustively across the entire validation set **three separate times** (once for each prompt type) to test ultimate **cross-modal robustness**.



### Text Prompts (GroundedSAM Approach)

- **Architecture:** Used a **frozen, pre-trained Grounding DINO** model to generate bounding boxes from the text prompt: "Person. Car"
- **Training Challenge (Alignment):** DINO generates a **variable number of boxes**, breaking the strict 1:1 mapping required to compute loss against GT masks.
- **Training Solution:** Mapped DINO boxes to GT masks using **maximum IoU matching** ( $\text{IoU} > 0.5$ ). DINO boxes that didn't match a GT object were heavily penalized by pairing them with **empty (zero) masks**.
- **Validation (Zero-Leakage):** Passed **all DINO-generated boxes** directly to SAM. GT boxes were **strictly ignored** to evaluate true end-to-end text-to-mask performance.

# Task e) Augmentation Strategy

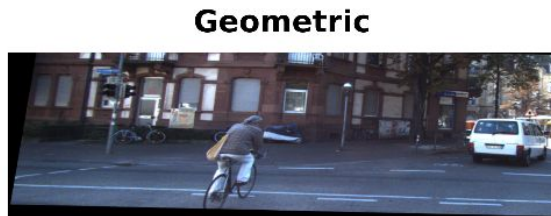
Augmentation strategies tried for the **bbbox prompt** model (same as Week1)



Original image  
No augmentations applied



Standard baseline  
Horizontal Flip (50%)



Spatial transformations  
+ Shift, Scale, Rotate, Perspective



Lighting & color shifts  
+ Brightness, Contrast, Gamma, RGB



Environmental hazards  
+ Rain, Fog, Shadows, or Sun Flare



Sensor & lens artifacts  
+ Noise, Blur, Dropout, JPEG Artifacts

## Legacy



Chaotic mix of all early tests

## Limit Test



All transformations compounded

# Task e) Augmentation Search

## Results & Reasoning

- **Peak Performance:** All fine-tuned models beat the pretrained, but strictly **No Augmentation** achieved the highest mAP.
- **Reasoning (Domain Homogeneity):** The dataset is clean and visually consistent across all splits. Injecting artificial distortions and sifts forces the model to learn **irrelevant noise**, actually harming performance.
- **Strategy Locked:** Due to this domain consistency and time constraints, we confidently fixed the **No Aug** strategy for all subsequent prompt experiments.

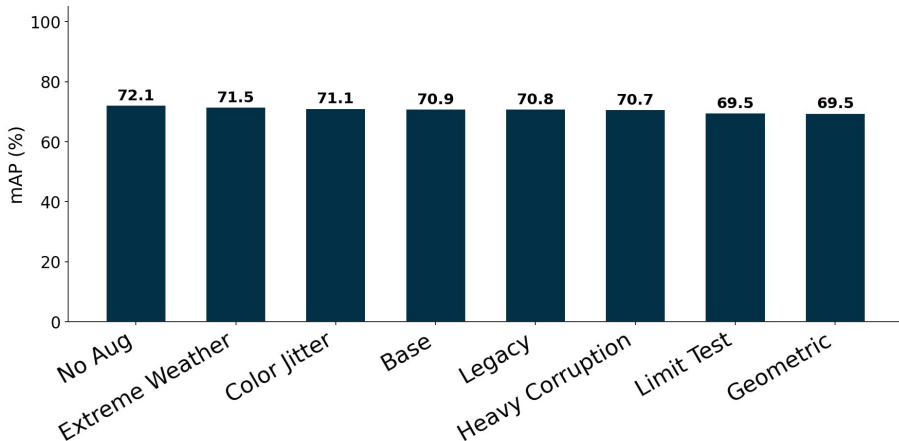
## Decision

- **Selected Pipeline:** No Augmentation
- **Dev mAP:** **72.1**
- **Improvement:** **+4.5 mAP**

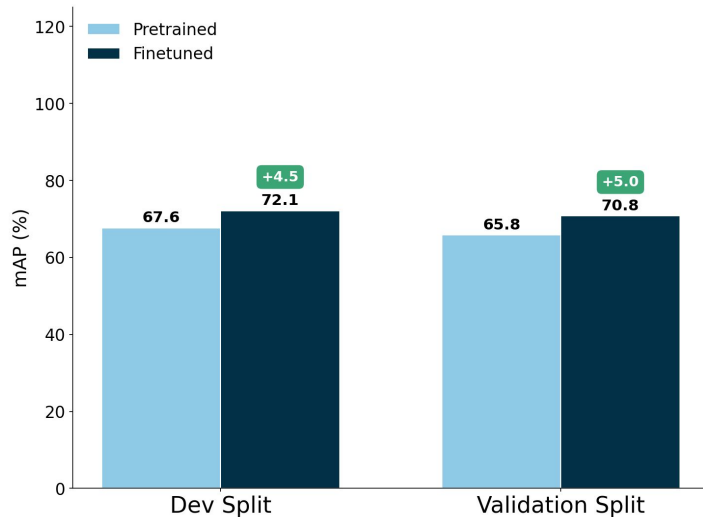
## Generalization

- **Tuned (Train/Dev):** **72.1 mAP**
- **Final (Full Train/Val):** **70.8 mAP**
- **Acceptable Gap:** A small, expected **generalization gap (-1.3 mAP)** when transitioning to unseen validation data.

Augmentation Strategy Comparison - mAP



Finetuning Generalization - Dev vs. Validation Splits

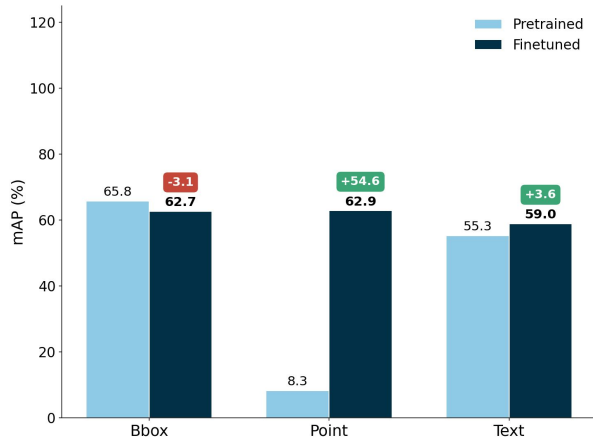


# Task e) Quantitative Evaluation

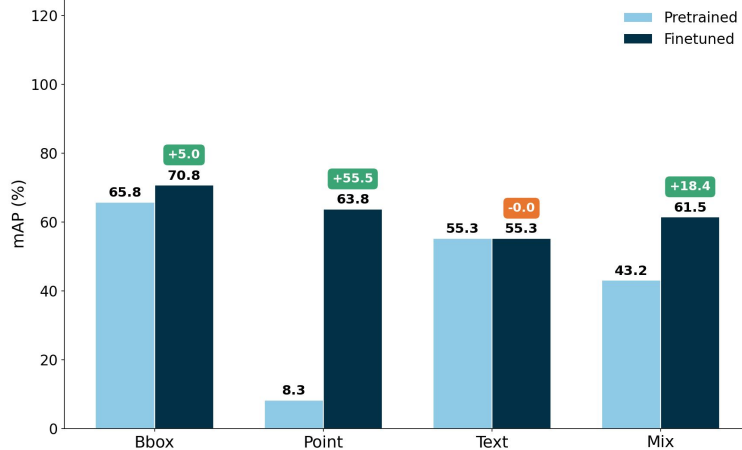
## Overall mAP per Input Type

- **Point Prompt Increase**: Pre-trained SAM leans toward **segmenting sub-parts** (e.g., just a car door). Fine-tuning successfully teaches the decoder to **infer the entire object** from a single center point.
- **Text Prompt Bottleneck**: The **frozen Grounding DINO** acts as a **hard ceiling**. If DINO produces false-positives or misses objects, the **SAM mask decoder cannot recover the lost mAP**, regardless of fine-tuning.
- **Bbox Ceiling**: Bounding boxes provide the **highest structural certainty**. The pre-trained baseline was **already near optimal**, so fine-tuning yields only marginal gains.

Mix Model - mAP per Sub-Prompt Type (Pre vs. Finetuned)



Overall mAP - Pretrained vs. Finetuned per Input Type



## mAP per Sub-Prompt Type (Mix Model)

- **Cross-Modal Synergy (Text)**: Surprisingly, the Mix model handles Text prompts better than the specialized Text model. **Reasoning**: Mixing precise (GT) and noisy (DINO) inputs during training **regularizes the decoder**, making it **more robust to imperfect boxes**.
- **Modality Trade-off (Bbox)**: The Mix model slightly regresses (-3.1 mAP) on Bbox prompts. This reflects a "**capacity tax**": the model **sacrifices peak performance** on explicit prompts to **build shared robustness** for ambiguous ones (like points).



# Task e) Quantitative Evaluation

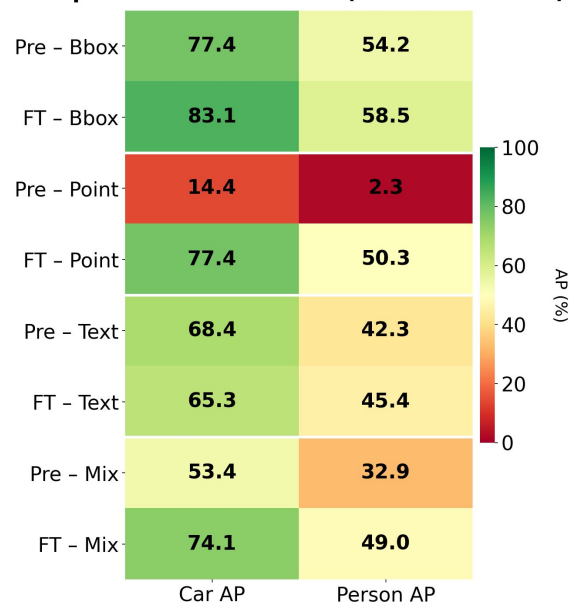
## AP per Class: Cars vs. Pedestrians

- **Articulation Penalty:** Person AP consistently lags behind Car AP across all inputs.  
**Reasoning:** Humans have **highly articulated poses**, demanding tighter pixel-level precision than rigid, predictable cars.
- **Text Bottleneck:** Text is the only modality showing just a **marginal increase or slight decrease** per class after fine-tuning.  
**Reasoning:** Frozen DINO boxes **mismatch GT annotations**. SAM's fine-tuning cannot fix these upstream localization errors.

mAP by Object Size - All Models (Pre & Finetuned)



AP per Class - All Models (Pre & Finetuned)



## mAP by Object Size

- **Point-Scale Anomaly:** Surprisingly, the pre-trained point model performs better on small objects than large ones.  
**Reasoning:** On large objects, a single point triggers a **sub-part mask** (e.g., just a door). On small objects, the point naturally covers the **entire object footprint**.
- **Context Overcomes Scale:** Fine-tuning fixes this anomaly; larger sizes now correctly yield higher mAP.  
**Reasoning:** Once the decoder is taught to segment the "whole" object, **larger sizes provide more spatial context**, enabling robust and precise boundaries.

# Task e) Quantitative Evaluation

## Radar Finetuned Models Summary

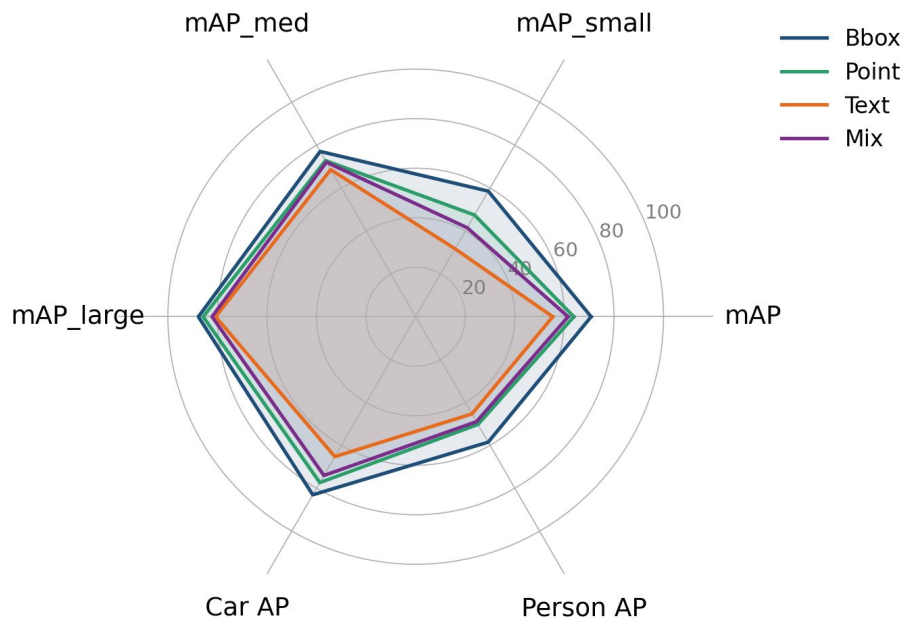
### Overall Modality Comparison

- Spatial Precision Hierarchy:** Performance directly correlates with input certainty (*Bbox* → *Point* → *Mix* → *Text*).  
**Reasoning:** GT Bboxes provide **absolute structural boundaries**, while Text suffers from **upstream DINO localization errors**.
- The Generalist Compromise:** The Mix model balances performance across the board.  
**Reasoning:** Training on diverse prompts forces a trade-off: **sacrificing peak Bbox accuracy to maintain robustness on ambiguous inputs**.

### Vulnerability at Small Scales (mAP\_small)

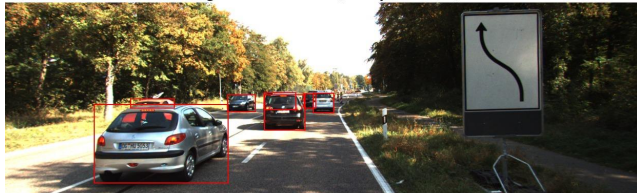
- The Scale Gap:** Modality differences are most extreme on small objects.  
**Reasoning:** Low pixel-counts make small targets **highly intolerant to prompt ambiguity** (e.g., a slightly misaligned Text box heavily degrades the mask).

## Finetuned Models - Radar Comparison



# Task e) Bbox prompt: *Qualitative Evaluation*

Original + GT BBoxes (Prompt: BBoxes)



Ground Truth Masks



Pretrained SAM



Finetuned SAM



## Resolving Rigid Occlusions

- **Pre-trained:** Overlapping boxes fracture the front car's mask.
  - **Fine-tuned:** Perfectly separates both cars despite severe box overlap.
- The **decoder successfully learned** to untangle depth and continuous boundaries for predictable, rigid shapes.

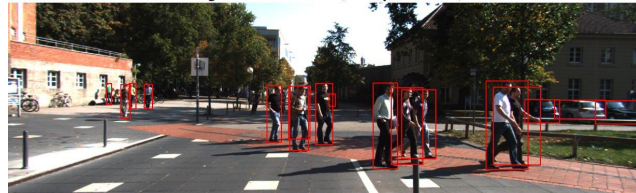


## Severe Articulated Occlusions

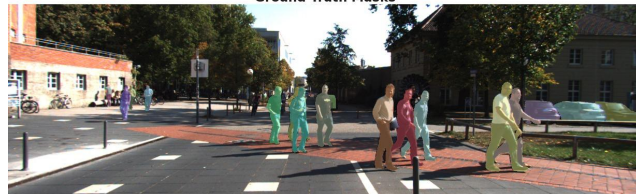
- Both models **fail** to cleanly **separate crossing pedestrians** (fine-tuning worsens fragmentation).
- Overlapping body parts create nearly identical boxes. Without rigid geometry, the **decoder struggles** to assign pixels to the correct overlapping prompt.



Original + GT BBoxes (Prompt: BBoxes)



Ground Truth Masks



Pretrained SAM



Finetuned SAM





# Task e) Bbox prompt: *Qualitative Evaluation*

Original + GT BBoxes (Prompt: BBoxes)



Ground Truth Masks



Pretrained SAM



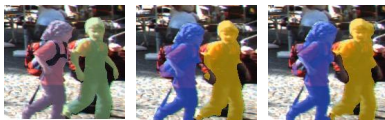
Finetuned SAM



## Fine Accessory Details

Both the pre-trained and fine-tuned models fail to segment **thin, attached features** like backpack straps.

➤ Bboxes provide macro-level boundaries but lack the **pixel-level guidance** needed to force the decoder to capture thin, ambiguous structures against complex backgrounds.



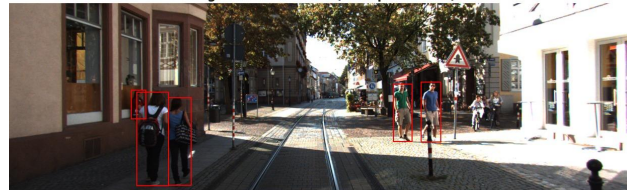
## Semantic Object Separation

Both models successfully **exclude some distinct handheld accessories** (like large bags) from the pedestrian mask.

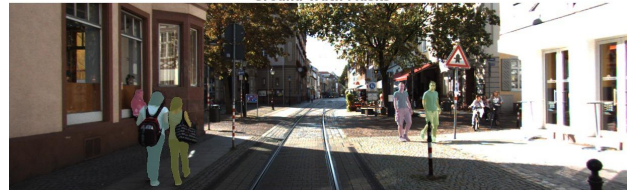
➤ The models effectively recognize the **core structural geometry** of a human body, naturally segmenting it away from **adjacent non-body objects** even if they fall perfectly inside the prompt box.



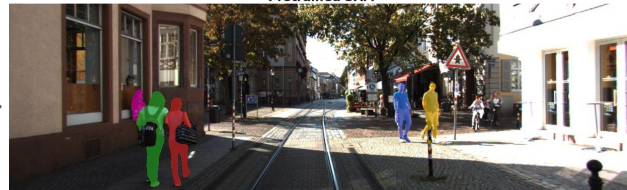
Original + GT BBoxes (Prompt: BBoxes)



Ground Truth Masks



Pretrained SAM



Finetuned SAM





# Task e) Point prompt: *Qualitative Evaluation*

Original + GT BBoxes (Prompt: Points)



Ground Truth Masks



Pretrained SAM



Finetuned SAM



## Whole-Object Segmentation

Fine-tuning successfully breaks the pre-trained model's bias for sub-parts with point prompts, teaching it to segment the **entire object**.

## Overcoming Part-Bias

Pre-trained SAM **Hyper-focuses on local structures** (e.g., just the window or license plate).

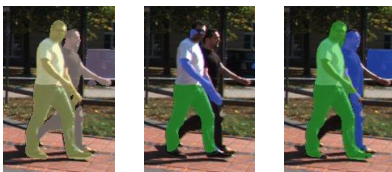
→ Fine-tuning forces the decoder to capture the **full macro-boundary**, ignoring localized textures.

## Occlusion Advantage

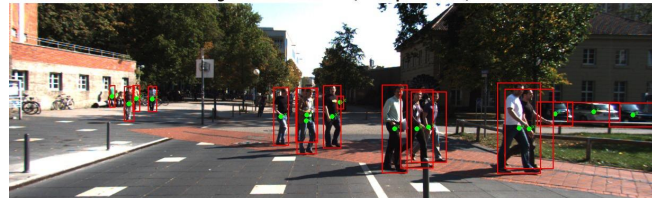
Pre-trained SAM isolates individual clothing items (shirts, pants).

Fine-tuning captures the whole person.

→ Because **point prompts do not physically overlap** (unlike boxes), the fine-tuned model can cleanly separate highly occluded pedestrians.



Original + GT BBoxes (Prompt: Points)



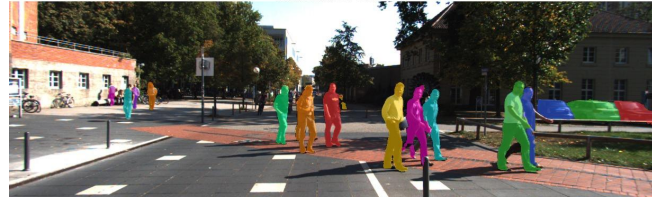
Ground Truth Masks



Pretrained SAM

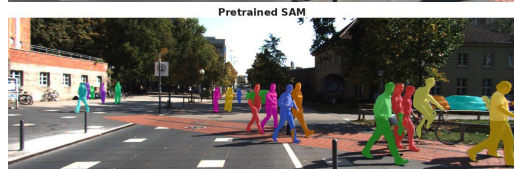
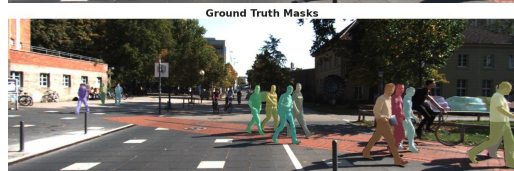
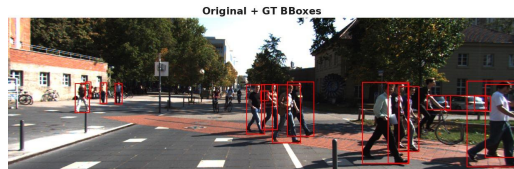


Finetuned SAM





# Task e) Text prompt: *Qualitative Evaluation*



## Learning Dataset Biases

GroundedSAM acts as a zero-shot detector, generating many more bounding boxes than the Ground Truth annotations.

- Fine-tuning teaches the decoder to act as a **strict dataset filter**, actively ignoring valid prompt boxes if they don't fit the original labeling rules.

## Class Suppression (Cyclists)

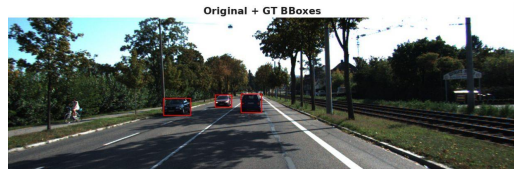
GroundedSAM correctly boxes cyclists in both images, but the fine-tuned model outputs an empty mask.

- Because KITTI-MOTS focuses strictly on pure pedestrians and cars, the model learned to **suppress unannotated edge-classes**, completely ignoring the explicit prompt.

## Size Penalty (Distant Objects)

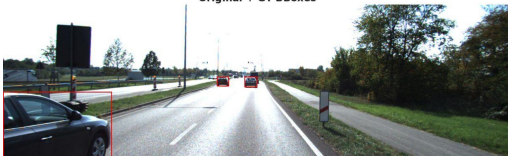
In the right image, a tiny distant car receives a prompt box but is ignored after fine-tuning.

- The decoder learned a **size-based threshold**. It rejects perfectly valid small boxes because the dataset's human annotators typically skipped them.

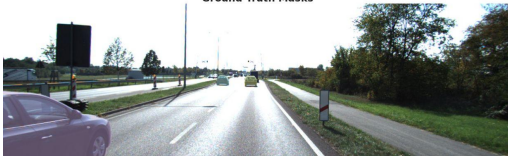


# Task e) Text prompt: *Qualitative Evaluation*

Original + GT BBoxes



Ground Truth Masks



Input BBoxes from GroundedSAM (Prompt: Text 'Person, Car')



Pretrained SAM



Finetuned SAM



## Filtering Non-Target Classes

Grounding DINO blindly detects traffic lights and tiny background cars, which the pre-trained model obediently segments.

✦ Fine-tuning successfully teaches the decoder to **suppress non-target objects**, outputting a clean mask that strictly adheres to the Ground Truth rules.



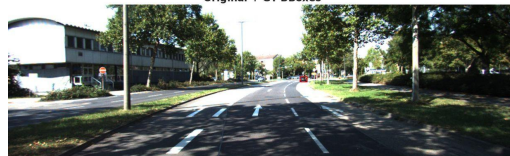
## Size Penalty (Distant Objects)

Two distant cars are similar in size (one is in the GT, one is not). Pre-trained segments both; fine-tuned **ignores both**.

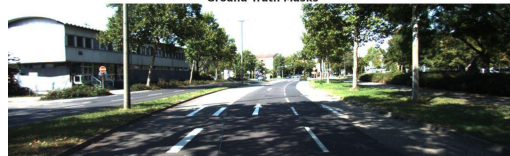
✦ By learning a **general rule to ignore tiny objects** (to reduce background noise), the fine-tuned model becomes **over-aggressive**, accidentally suppressing borderline small targets that are actually valid.



Original + GT BBoxes



Ground Truth Masks



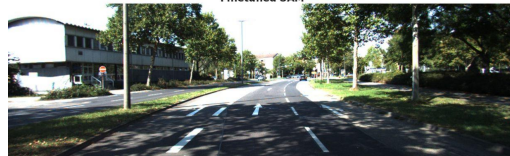
Input BBoxes from GroundedSAM (Prompt: Text 'Person, Car')



Pretrained SAM



Finetuned SAM





# Task f) DEArt: Domain shift

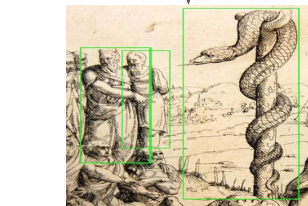
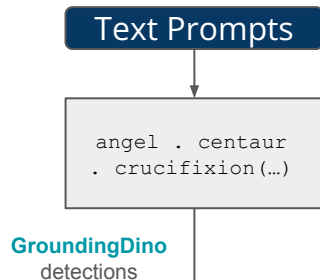
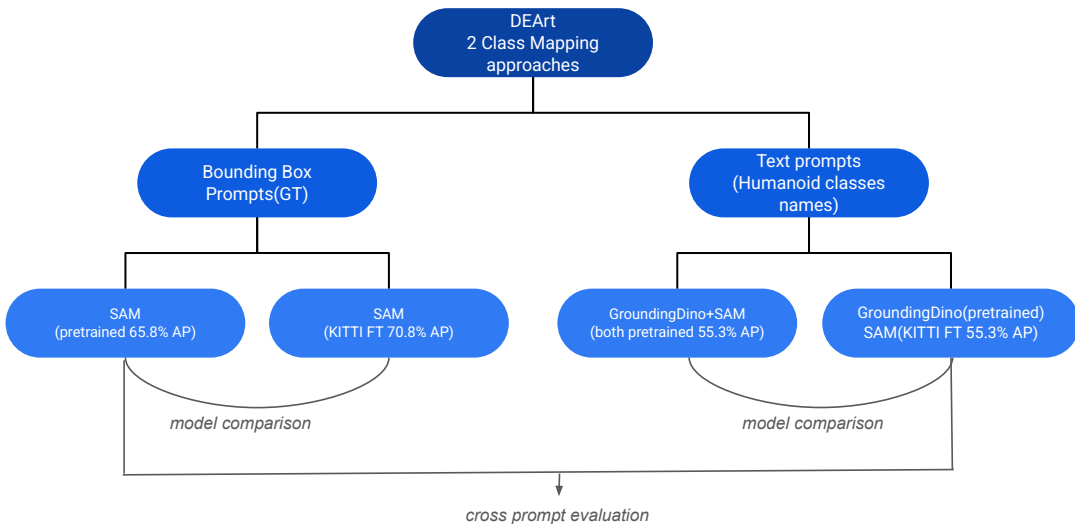
**Humanoid annotations:** 61,782 (52% out of total), **11 classes**.

**Masks:** **no** GT masks, used just bounding boxes

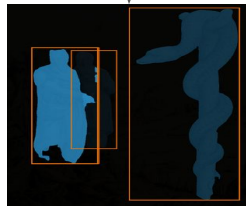
**Class map:**

- all **11 classes** mapped to person
- Unique class ID to see where the model fails the most

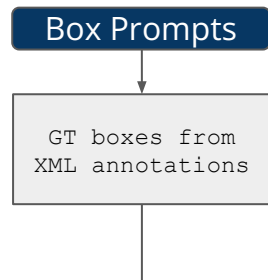
## Evaluation pipeline



**SAM masks**  
from **DINO** boxes



**Detected labels:**  
monk, person, crucifixion



**SAM masks**  
from **GT** boxes



**Isolated segmentation quality**

## Task f) DEArt: Metrics

The original objective was just to visually assess the models robustness to domain shift. However, to provide a more rigorous analysis, we tried to introduce a **quantitative** methodology. In the **absence** of ground-truth masks, we utilized (**created**) **two specific metrics** as reliable approximations for segmentation **quality, corresponding to the two models we evaluated** (SAM and Grounded SAM):

### Mask → Box IoU (SAM)

**Def:** Measures the IoU between the predicted mask and the filled GT bounding box.

**Why:** Acts as a proxy to evaluate segmentation accuracy since DEArt lacks true GT masks

$$IoU = \frac{\sum_{x,y} (P_{x,y} \cap B_{x,y})}{\sum_{x,y} (P_{x,y} \cup B_{x,y})}$$

where

\*P is the predicted binary mask.

\*B is the proxy mask (filled GT box).

Bounding boxes prompts\*

Image index: 0 | class\_id: 10 | IoU = 0.6389



Original image

GT Box Mask  
(we created)

Predicted mask

Intersection

### Bbox Count (GSam)

**Def:** Measures the difference between predicted and ground-truth bounding box counts per image.

**Why:** Quantifies domain shift by revealing if the model hallucinates objects (over-detection) or misses them (under-detection).

$$\Delta N = N_{det} - N_{gt}$$

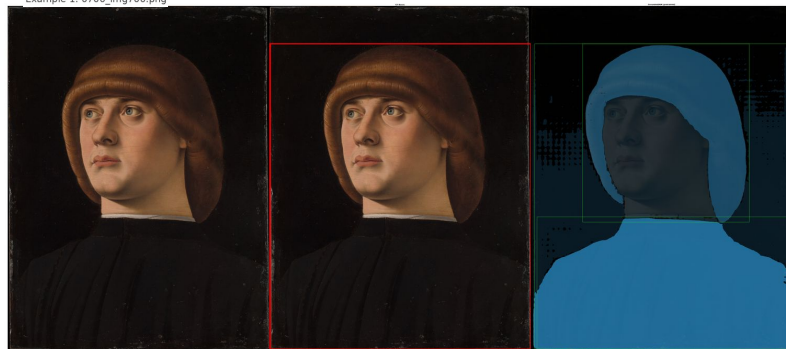
where

$N_{det}$  = number of predicted boxes.

$N_{gt}$  = number of ground-truth boxes.

Text prompts\*

Example 1: 0700\_img700.png



Original image

GT Bounding Box

Grounding DINO boxes  
+ SAM segmentation



## Task f) DEArt: Box prompts

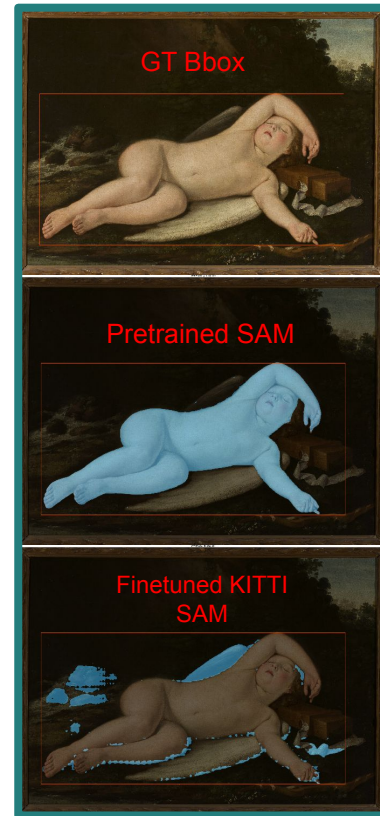
### Conclusions

- **GT Prompts Isolate Pure Segmentation:** By using the ground-truth boxes, the **detection errors** are eliminated entirely. In this isolated test, the pre-trained zero-shot model has strong object integrity, generating dense, **cohesive masks** for painted figures.
- **Fine-Tuning Causes Fragmentation:** Training on a unrelated domain (KITTI driving scenes) destroys the model's general shape priors, like in *Week 1*. When needed to segment art, the fine-tuned model outputs **broken** and **pixelated** mask fragments instead of solid shapes(in some examples).
- **IoU Drop:** The scattered nature of the fine-tuned masks helps us understand the quantitative decline. Because the broken pixels fail to fill the bounding box area, the **mask\_box\_iou** drops from 0.51 to 0.48, which confirm that the models are similar in segmentation and we cherry-picked the extreme cases.

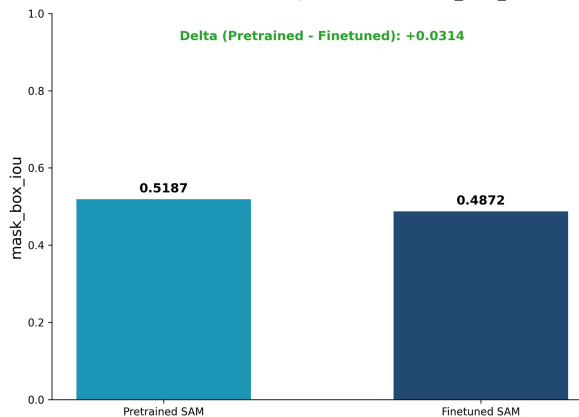
Image 745



Image 966



DeART GT-Box Prompt: Overall mask\_box\_iou



# Task f) DEArt: Text prompts

(Additionally, the fine-tuned SAM still exhibits the same severe pixel fragmentation seen previously)

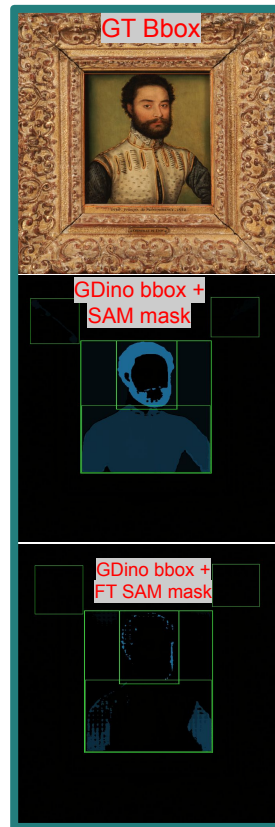
## Conclusions

- **Over-Detection (Domain Gap):** GroundingDINO predicts ~35% more boxes than the GT. Sometimes it **hallucinates** "figures" in ambiguous brushstrokes or background elements (like the background statues in *Image 374*) that **annotators ignored**.
- **Identical Box Counts Across Pipelines:** Both the pre-trained and FT pipelines detect the exact same number of bboxes. This is because we only fine-tuned SAM's mask decoder, the GroundingDINO object detector remained **completely frozen**, feeding the exact same boxes to both SAM models.
- **Flawed Prompts Degrade Segmentation:** Overall mask quality is **visibly worse** here than in the GT-box experiment. SAM is highly sensitive to prompt quality; because GroundingDINO's predicted boxes are often loose, off-center, or capture background noise, SAM is forced **to segment** those mistakes.

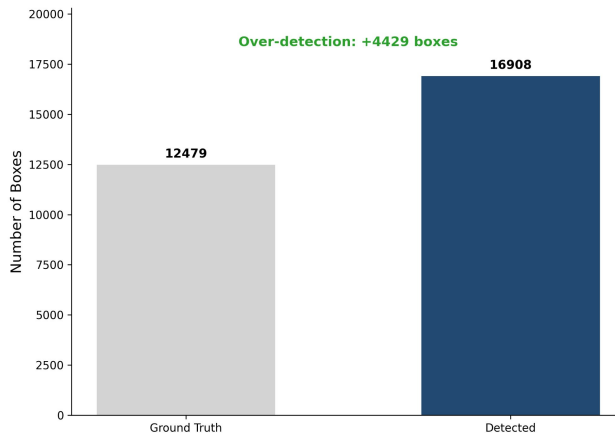
Image 374



Image 1077

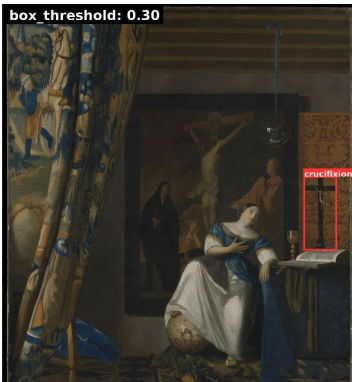
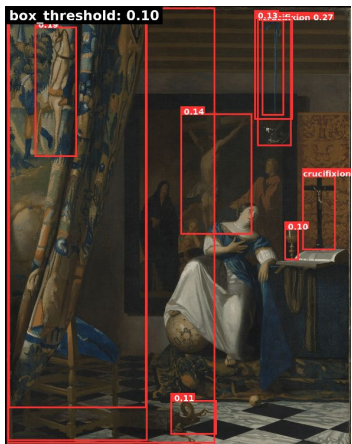


GroundingDINO Box Counts on DeART



# Task f) DEArt: Future work

*\*that we thought, but didn't try  
because of the time constraints*



## Fine-Tune the Object Detector

Training the detector(**GroundingDINO**) directly on the DEArt bounding boxes would teach it the correct visual semantics and context for these specific artistic classes, reducing false positives and thus over-detection.

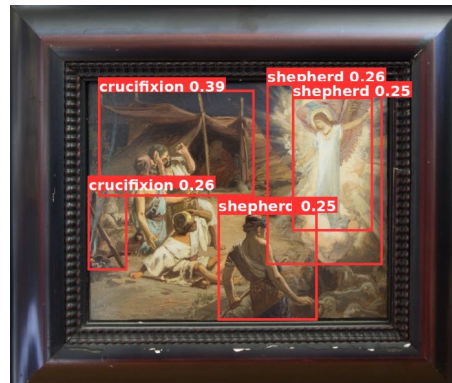
## Prompt Optimization

We wanted to play with **box\_thresh** and **text\_thresh**. Raising these thresholds would filter out the low-confidence noise or we could have used **non-max suppression**. Moreover, simplifying the text prompts, like just prompting for "person" and classifying them later, might confuse the detector less.

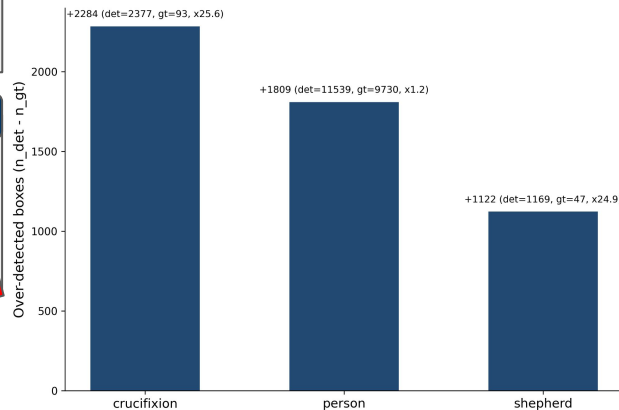
## Out-of-Distribution Hallucinations

Classes like **crucifixion** (25x over-detection) likely suffer from a massive domain gap. Because these specific concepts are **rare** in GDINO's pre-training data, the model likely over-triggers on generic shapes—identifying any cross-like structure (swords, frames, masts) as a "crucifixion".

**Just like in Week 1, the domain shift here is incredibly aggressive, leading to clear negative transfer!**



**Top-3 Most Over-Detected Classes (GroundingDINO)**



# Task g) Comparative Analysis Between Prompts *(Inference)*

| Model Inference Input               | Params (M)   | GFLOPs       | Peak VRAM (MB) | Latency (ms)  | FPS (Batch = 1) | Total Time Inference (s) |
|-------------------------------------|--------------|--------------|----------------|---------------|-----------------|--------------------------|
| <b>Point - Grid</b>                 | 93.74        | 746.4        | 3593.82        | 1115.08       | 0.90            | 3324.06                  |
| <b>Point - SIFT</b>                 | 93.74        | 746.4        | 2310.05        | 749.66        | 1.33            | 2234.73                  |
| <b>Point - Center GT BB</b>         | <b>93.74</b> | <b>746.4</b> | <b>1977.71</b> | <b>114.29</b> | <b>6.93</b>     | <b>430.12</b>            |
| <b>Text (Person. Car)</b> *         | 93.74 + 172  | 746.4 + 464  | 2637.74        | 280.66        | 3.56            | 836.66                   |
| <b>Bounding Box (FT detector)</b> * | 93.74 + 19.1 | 746.4 + 92   | 2077.25        | 165.96        | 6.03            | 494.74                   |

## Experimental Context

This table compares the **inference efficiency** of the prompting methods explored in sections [a\)](#), [b\)](#), and [c\)](#), evaluated using their **default parameters**.

**\* Note:** The + values in Parameters and GFLOPs indicate the added computational cost of the first-stage object detectors (Grounding DINO or YOLOv10-B) required to generate the boxes for SAM.

## Key Observations

- **The Prompt Bottleneck:** SAM's inference speed is heavily dictated by the **quantity of input prompts**. Sparse, precise inputs are significantly faster than dense ones.
- **Most Efficient (Point - Center GT BB):** The **fastest approach**. It provides SAM with **minimal inputs** and requires zero external detection overhead.
- **Least Efficient (Point - Grid):** The **slowest approach**. Generating a **dense grid** floods SAM with points, maximizing processing time.
- **The SIFT Middle Ground:** Processes a **fixed, moderate number** of keypoints, placing its efficiency squarely between the single-point and dense-grid methods.
- **Two-Stage Overhead** (Text vs. BBox): Both require an initial model to generate boxes for SAM, but their efficiency differs:
  - **BBox (FT Detector):** **Highly efficient** because the fine-tuned YOLOv10-B detector is extremely lightweight.
  - **Text (Person. Car):** **Noticeably slower** because the Grounding DINO model is **computationally heavy** and generally produces a **higher volume of box** candidates.



# Task g) Comparative Analysis Between Prompts (FT)

| Model Finetune Input | Total Runtime (15 epochs) | Avg s/epoch           | Validation mAP | Validation F1@50 |
|----------------------|---------------------------|-----------------------|----------------|------------------|
| <b>Bounding Box</b>  | <b>2h 25m 15s</b>         | <b>~581s (~9m41s)</b> | <b>70.8</b>    | <b>0.9548</b>    |
| <b>Point</b>         | <b>2h 23m 40s</b>         | <b>~574s (~9m34s)</b> | 63.8           | 0.9048           |
| <b>Text</b>          | 6h 46m 50s                | ~1627s (~27m07s)      | 55.3           | 0.7683           |
| <b>Mix</b>           | 6h 19m 14s                | ~1517s (~25m17s)      | 61.5           | 0.8609           |

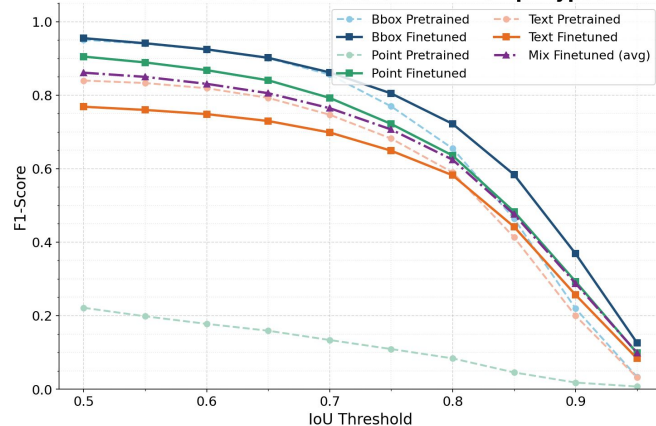
## Experimental Context

This section evaluates the **training efficiency** and **segmentation quality** of the fine-tuned models from section [e\)](#). We introduce the **F1-Score** alongside our primary mAP metric to provide a more comprehensive analysis of model **robustness** across varying IoU thresholds.

## Key Observations

- **Training Efficiency:** **BBox** and **Point** train quickly and similarly. **Text** and **Mix** are significantly slower due to the **computational overhead** of running the first-stage detector (Grounding DINO) during training.
- **Peak Performance (BBox):** Bounding boxes yield the highest validation mAP and F1, providing SAM with the most **stable and unambiguous input**.
- **Robustness to Strict IoU (Plot Insights):** As the evaluation IoU threshold becomes stricter, the F1-Score naturally degrades across all models.
- **Finetuned vs. Pretrained:** While the fine-tuned models show a slightly **steeper performance drop-off** at very high IoUs compared to the pre-trained baseline, they still **consistently outperform** the pre-trained model at every single threshold.
- **Consistent Ranking:** The models' F1-Score robustness perfectly mirrors their baseline mAP ranking (**Bbox** > **Point** > **Mix** > **Text**).

IoU Threshold vs F1-Score — All Prompt Types



# Task g) Qualitative Analysis Between Prompts (FT)

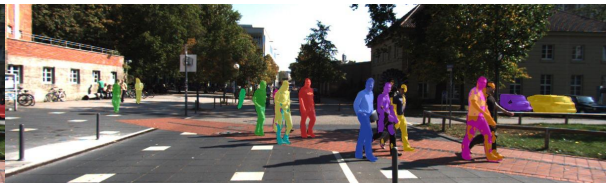
Bbox



Point



Text



## Experimental Context

Building on the baseline comparisons in section [e](#)), this slide isolates the **qualitative differences** between the fine-tuned models driven by **different prompting strategies** (Bounding Box, Point, and Text).

## Key Observations

- **Annotation Dependency** (*Top Row*): The **BBox** and **Point** models perfectly mirror the Ground Truth (GT) annotations they rely on. They **successfully segment** difficult, **overlapping cars** present in the GT, but **ignore a distant car** that was left **unannotated**. Conversely, the **Text** model (driven by Grounding DINO) operates **independently of the GT bias**, resulting in slightly **different, yet valid**, segmentation choices.
- **The Overlapping Box Failure** (*Bottom Row*): Prompts relying on bounding boxes (both strict BBox and Text) struggle significantly with **highly articulated, overlapping subjects** like crossing pedestrians. When bodies overlap, their bounding **boxes become nearly identical**, confusing SAM's decoder and causing **fragmented or merged masks**.
- **Point Prompt Superiority for Crowds**: Point prompts completely **bypass the overlapping box issue**. Because a single 1x1 pixel coordinate cannot "overlap" another, SAM clearly distinguishes between subjects, resulting in much **cleaner and highly precise** pedestrian segmentations.

## Task h) OPTIONAL: Preprocessing

### Training Preprocessing (Merging Instances)

- **How:** We merged overlapping instance masks of the same class into a single **binary matrix** using a bitwise OR operation.
- **Why:** To train SAM to predict an entire **semantic cluster** for a specific class, rather than just isolated individual objects



### Evaluation Preprocessing (Building Dense Maps)

- **How:** We applied class IDs directly to a blank matrix, processing overlapping instances sequentially to flatten them into a **single 2D map**.
- **Why:** To convert the annotations into true semantic dense-maps so we could evaluate the model using standard **COCO Semantic** metrics.



### Inference (Passing Labels to SAM)

- **How:** We fed the model ground-truth bounding boxes, injected the corresponding **class label** into the generated mask, and resolved overlaps by ranking SAM's internal confidence scores.
- **Why:** Because SAM only outputs binary masks without class names. This allowed us to link boundaries to classes and ensure the **highest confidence** prediction overwrites weaker ones.

1. Original Image (RGB)



2. Instance Annotations (Source Dataset)



3. Semantic Preprocessing (Target/Input)



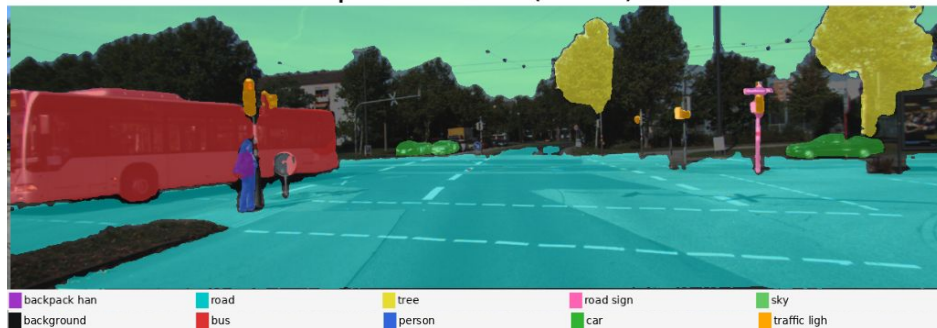
## Task h) OPTIONAL: Text prompts

For using **text queries** to detect the objects, we map the detected text labels to standard Class IDs using a custom dictionary, then flatten the masks onto a **2D semantic map**, resolving overlaps by prioritizing **high-confidence predictions**. While this approach naturally tends to **oversegment** the scene as we have seen in the previous tasks, the overall semantic segmentation remains **highly effective**. To explore this, we tried a 3-word prompt ("cars, person, background") and a rich, open-vocabulary approach.

### Expanded Vocabulary

- We wanted to see how many objects we can segment **without** the **needing of GT boxes**.
- **Vocabulary Used:** ≈80 COCO categories highly relevant to driving scenes, plus additional urban/outdoor concepts that **GroundingDINO** handles well.
- We consider it offers a **strong trade-off** between performance and the need for manual annotations or fine-tuning.

Open-Vocab Semantic (9 classes)



GT Semantic

Predicted Semantic (text\_prompt)



**Prompt:** Person . Cars. Background



# Task h) OPTIONAL: Finetuned semantic vs pretrained

## Aggressive Generalization

- The fine-tuned model becomes **hypersensitive** to features like vehicles. It aggressively segments objects like buses, vans, and heavily occluded background cars. While visually accurate, these classes were intentionally excluded from the KITTI GT, meaning the model is **penalized** for being "too good" at finding vehicles.

## Loss of Shape Priors

- By fine-tuning on driving scenes, the model **overfits** to local textures at the expense of true 3D geometry. Moreover, it loses the robust shape priors of the original SAM, misclassifying flat road markings like the painted directional arrow on the asphalt as a car. Though, the **semantic classes** are mostly correctly identified.

## Dataset Ambiguity

- We consider the comparison **not totally fair** because some false positives are frequently correct predictions. The fine-tuned model actually exposes the **flaws** and inconsistencies in the GT, showing valid vehicles that human annotators simply missed, ignored, or deemed too small.

GT Semantic

Predicted Semantic (pretrained\_sam)

Predicted Semantic (finetuned\_sam)



GT Semantic

Predicted Semantic (pretrained\_sam)

Predicted Semantic (finetuned\_sam)



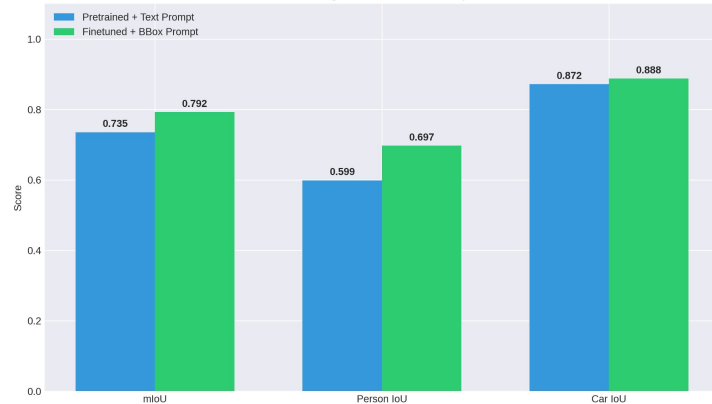
background car person

# Task h) OPTIONAL: Fine-tuned semantic vs

## Why Fine-Tuning Wins

- **Fine-Tuned Superiority:** The fine-tuned model **successfully** adapts to the specific visual context of the dataset (driving scenes) and has higher mIoU than the pretrained model using text prompts. It learns the exact scale, shapes, and typical locations of cars and pedestrians, resulting in **highly precise masks**.
- **Visual Proof:** In the text-prompt prediction, GroundingDINO **hallucinates** features such as misclassifying a vertical radio antenna as a "person" (red mask)—lowering its mIoU score. **1**

Semantic Segmentation Metrics Comparison



Original

GT Semantic

Predicted Semantic (text\_prompt)

Original

GT Semantic

Predicted Semantic (finetuned\_sam)



# Summary

GT Bounding  
Box



GT  
Segmentation



Text Bounding  
Box



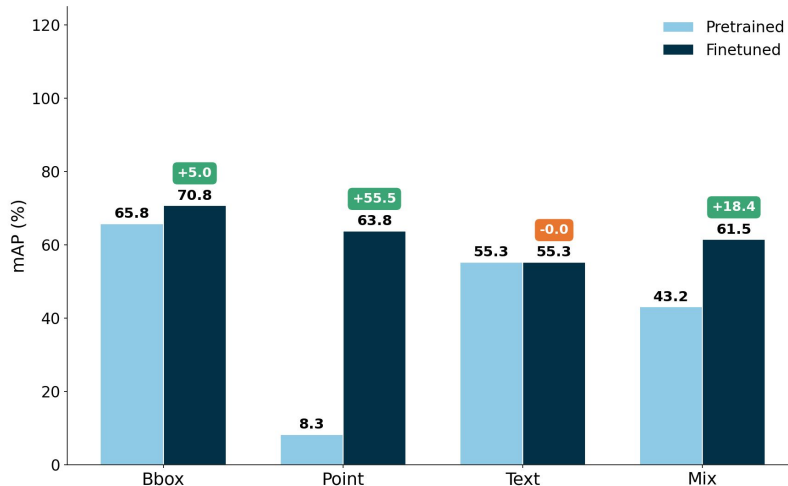
Pretrained  
Segmentation



Fine-Tuned  
Segmentation



Overall mAP - Pretrained vs. Finetuned per Input Type

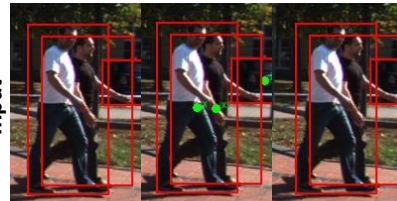


GT Bbox

Point

Text Bbox

Input



Segmentation



## Conclusions

- **Fixing *Part-Bias*:** Pretrained SAM often segments **fragmented sub-parts**. Fine-tuning forces the decoder to predict **complete, cohesive instances**.
- **Text Bottleneck:** Fine-tuning boosts **spatial prompts (Point/BBox)**. However, **Text prompts show zero improvement** because performance is **hard-capped** by the external object detector (**Grounding DINO**).
- **Beyond the Ground Truth:** The fine-tuned model exposes dataset flaws by finding valid vehicles annotators missed. Our true real-world performance is likely higher than measured mAP.
- **Resolving Crowd Overlap:** **Bounding boxes fail** in dense scenes because overlap confuses the model. **Point prompts bypass this**, providing distinct targets to cleanly separate packed subjects.
- **Aggressive Domain Shift (DeART):** Just like Week 1, the extreme visual shift to DeART leads to **clear negative transfer**.
- **Inference Bottlenecks:** Speed is dictated by **prompt volume** and **detector overhead**. Sparse **single-point prompts are fastest**, while **dense grids** and heavy models (Grounding DINO) cause **severe slowdowns**.

Point  
prompt



Pretrained  
Segmentation



Fine-Tuned  
Segmentation

